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Chapter 4

Ethereum Blockchain Basics

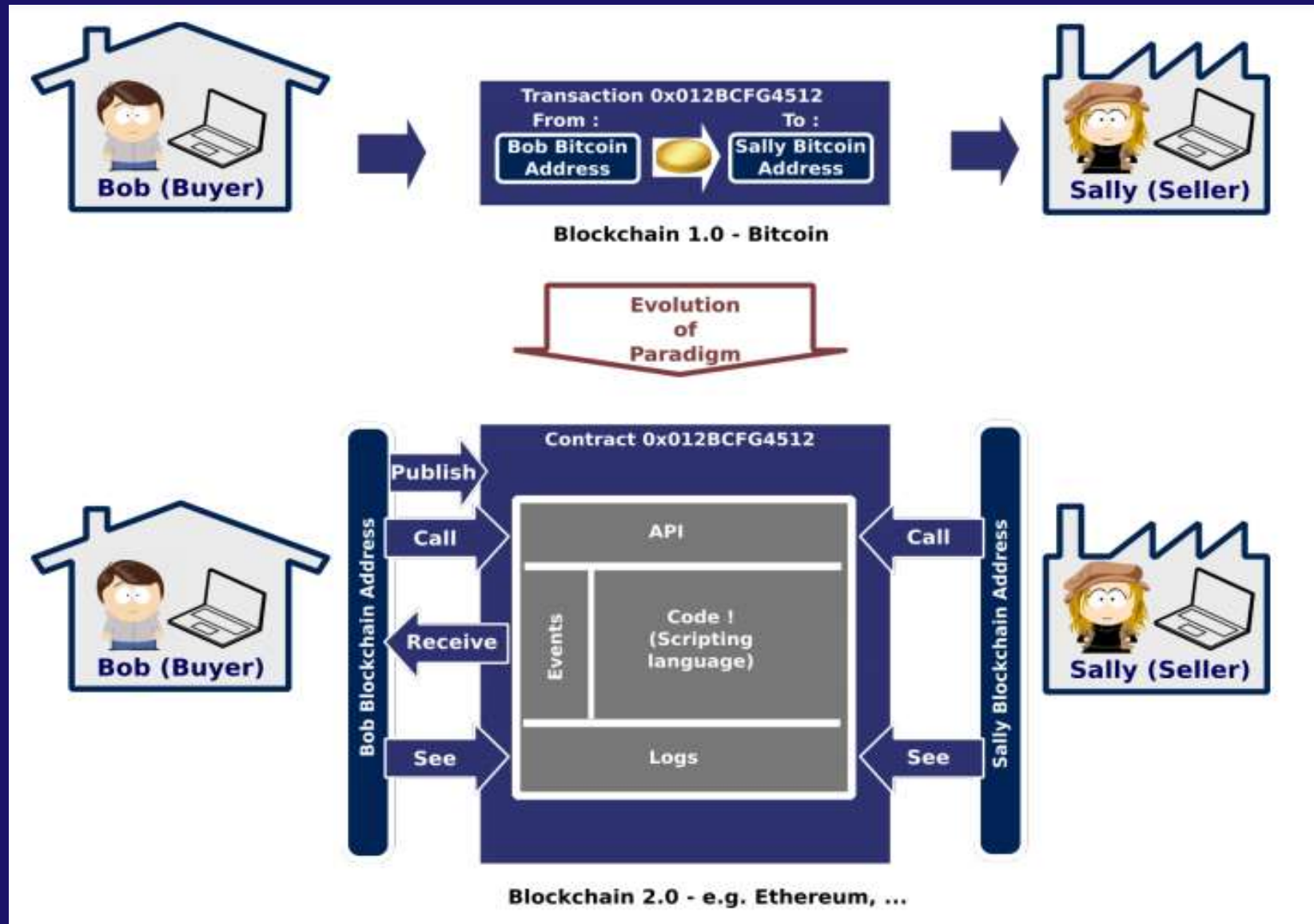
Ethereum

- Ethereum is a public blockchain, decentralized and distributed computing platform
- Proposed in late 2013 by Vitalik Buterin (cryptocurrency researcher and programmer)
- Not just money : Asset issuance, gambling, IoT, voting, supply chain, document verification, hundreds of applications
- Cryptocurrency : Ether (BTC fo BC)
- Ethereum Consensus : PoW (idem for BC)
- Hash function : Ethash (SHA 256 for BC)
- Blocktime : 14 seconds (10 minutes for BC)

Ethereum

- Ethereum platform is a general purpose Blockchain or programmable blockchain : **Generalized Technology**, in opposite to Bitcoin which is dedicated to manage cryptocurrency
- Ethereum is often described and interpreted as a distributed global computer “The **world computer**”
- Programmable blockchain
 - ↳ Allows users to develop their own decentralized applications (dApps)
- Ethereum provides support for Turing-complete language for application development
- Transactions record code, variables, the results of function calls, ...
- Transactions are generated by a computer program : **Smart Contract**
- Application can be written in several languages, ranging from general-purpose languages like Java, C++, Python, JavaScript, Golang to platform-specific languages like **Solidity** for Ethereum or Bitcoin Script for Bitcoin.

Ethereum



Ethereum

- For execution, Ethereum provides a trusted and decentralised virtual machine as runtime environment : Ethereum Virtual Machine (**EVM**).
- EVM is a stack based processor.

Let's summarize

Ethereum : Programmable blockchain platform (Open Source)

Cryptocurrency : Ether

Solidity : Programming language

Smart Contract : Program

EVM : Execution runtime for Smart Contract written in Solidity

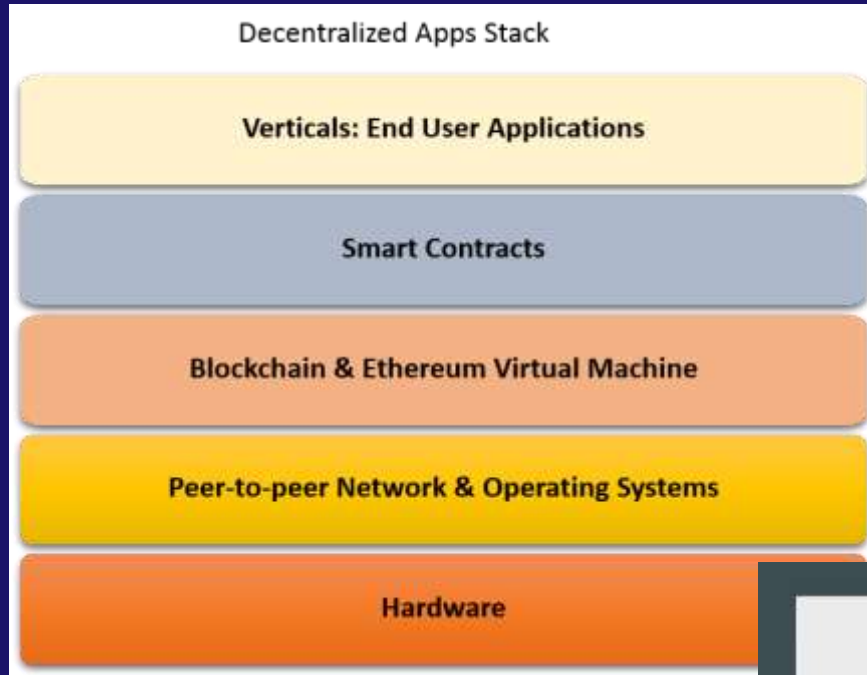
Compilation : Smart Contract is compiled into EVM assembly **Byte Code**

- To learn solidity go to **<https://remix.ethereum.org>** and you can start programming smart contracts without having to create your own Ethereum network platform.

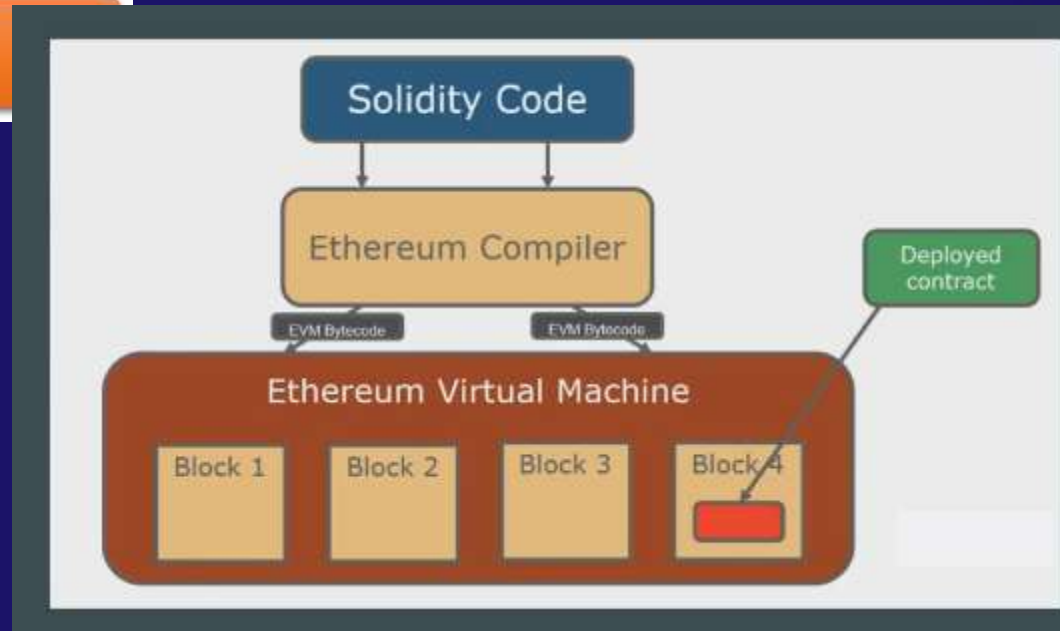
Ethereum : Ether units

Value (in wei)	Exponent	Common name	SI name
1	1	wei	Wei
1,000	10^3	Babbage	Kilowei or femtoether
1,000,000	10^6	Lovelace	Megawei or picoether
1,000,000,000	10^9	Shannon	Gigawei or nanoether
1,000,000,000,000	10^{12}	Szabo	Microether or micro
1,000,000,000,000,000	10^{15}	Finney	Milliether or milli
<i>1,000,000,000,000,000,000</i>	<i>10^{18}</i>	<i>Ether</i>	<i>Ether</i>
1,000,000,000,000,000,000,000	10^{21}	Grand	Kiloether
1,000,000,000,000,000,000,000,000	10^{24}		Megaether

Ethereum : Single Node

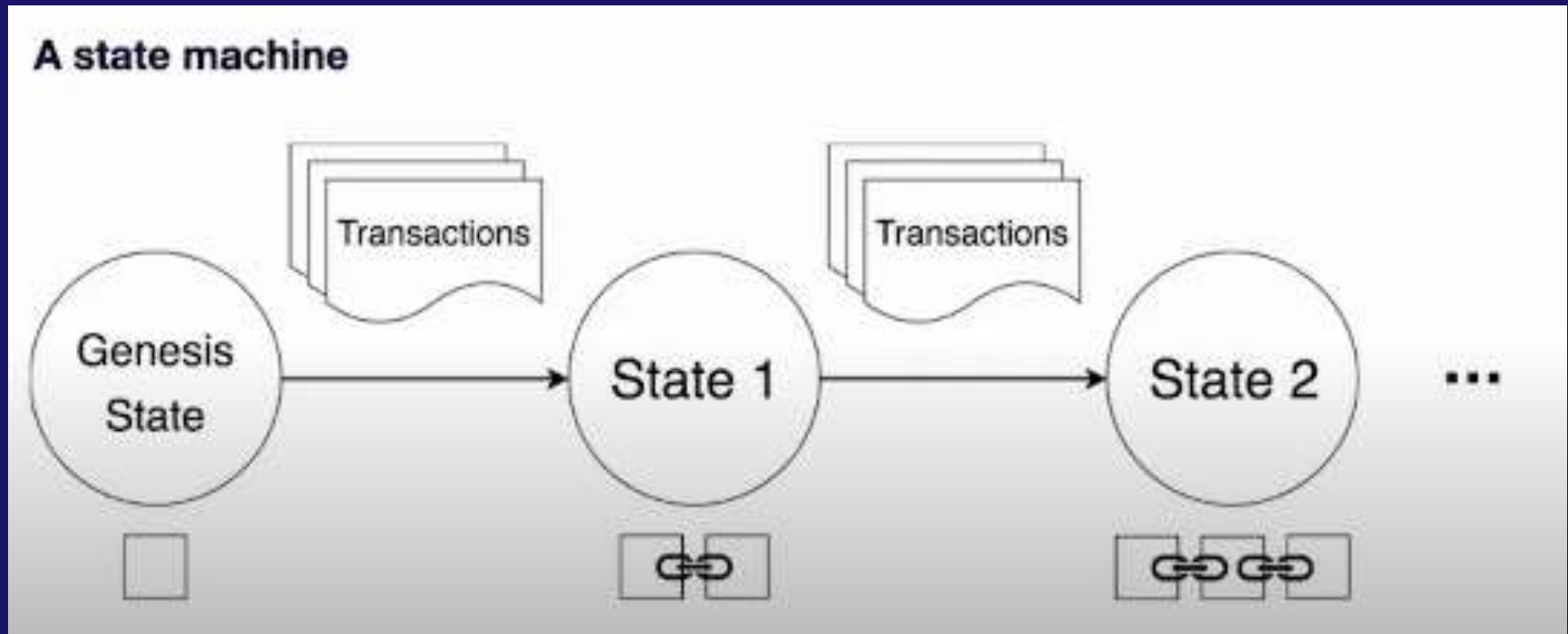


Web development : JS,
HTML, ...



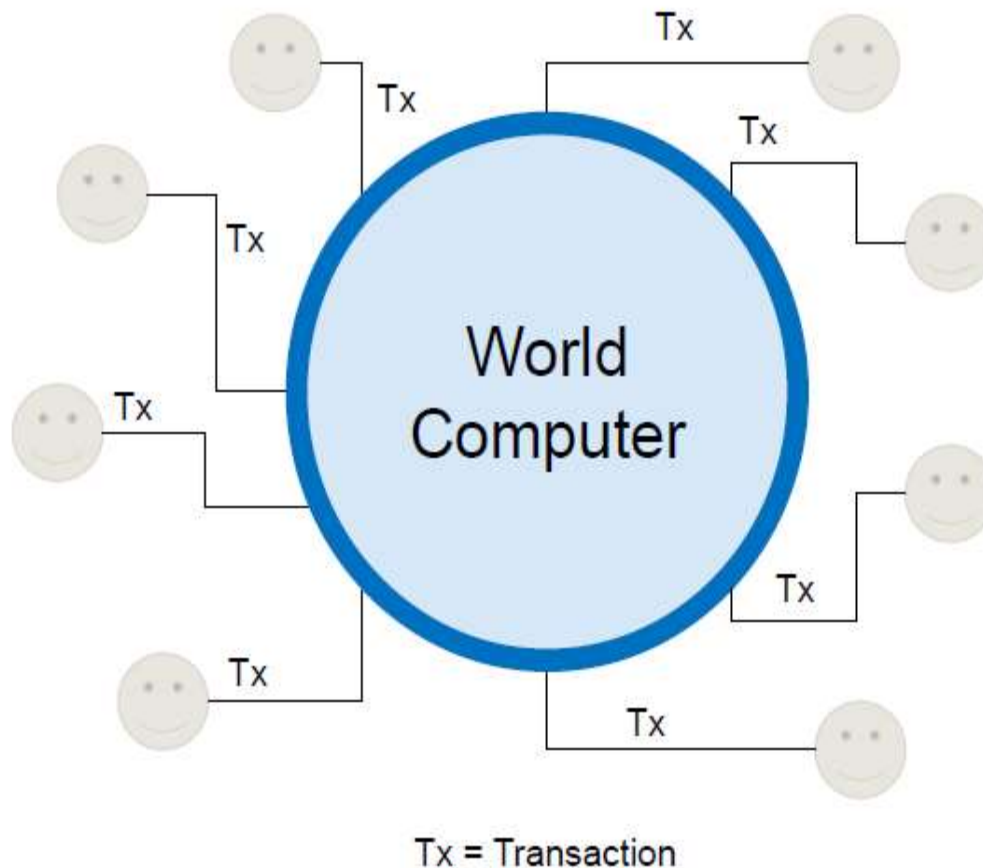
Ethereum

- From a computer science perspective, Ethereum is a deterministic but practically unbounded state machine, consisting of a globally accessible singleton state and a virtual machine that applies changes to that state
- Ethereum can be perceived as a distributed transaction-based state machine



The concept of a world computer

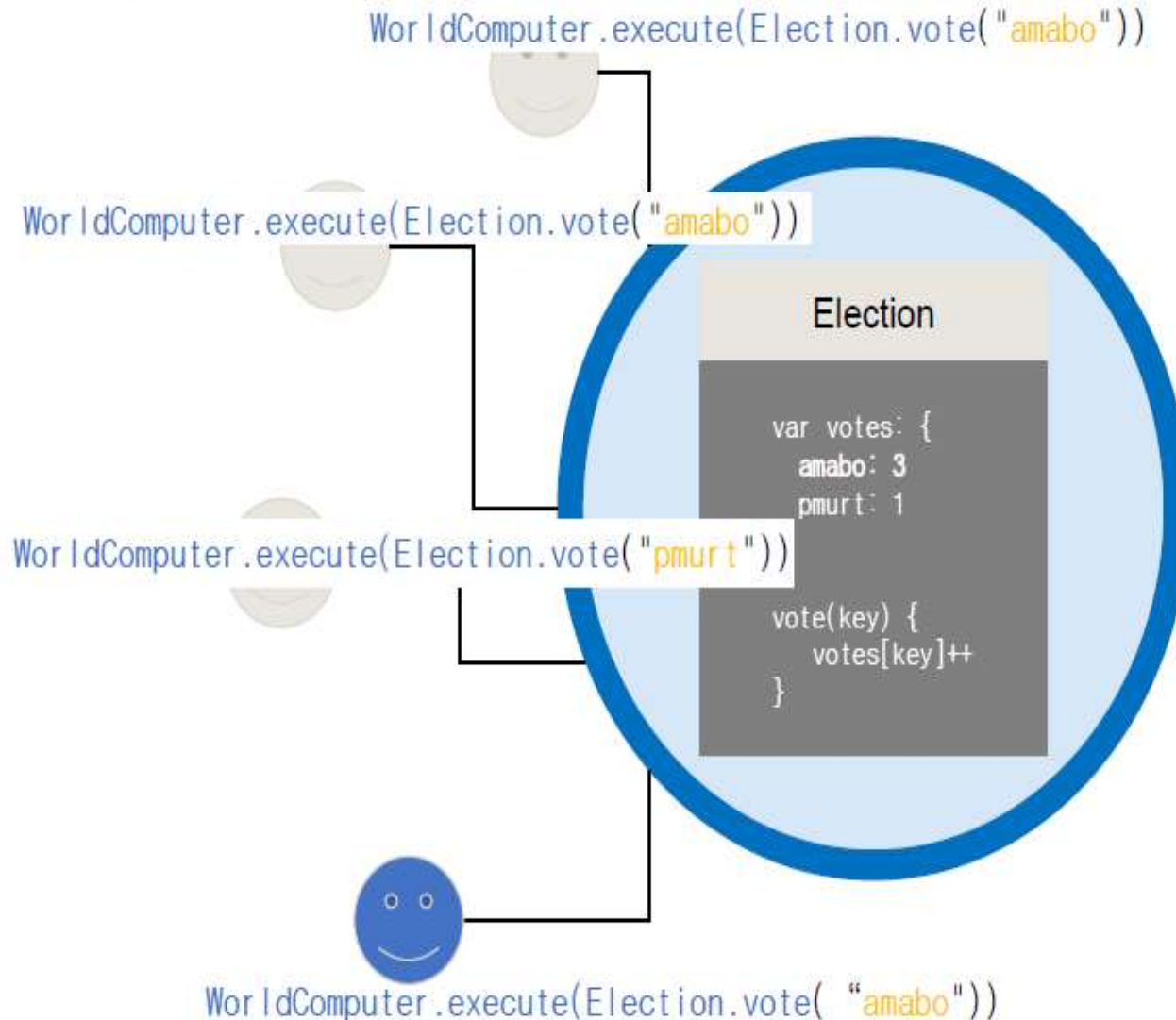
State Machine



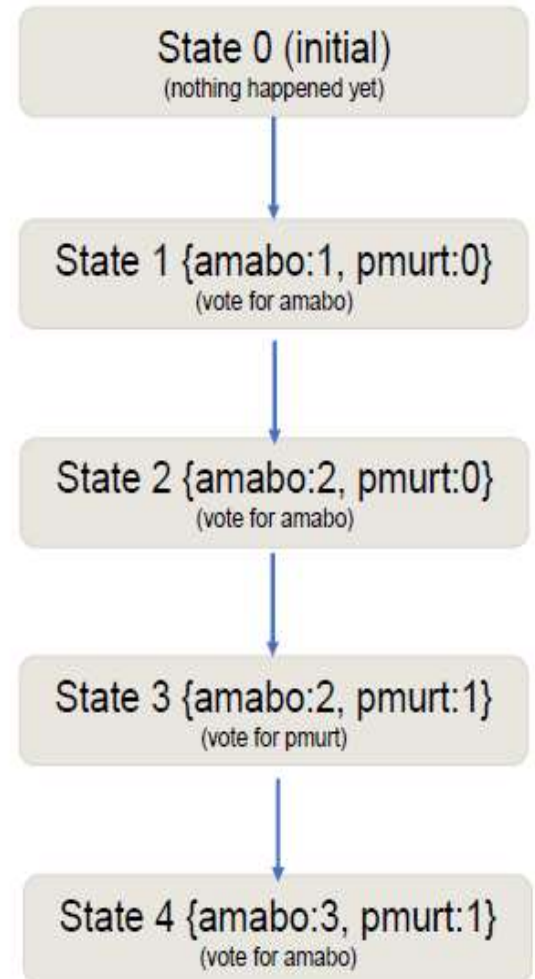
Properties

- All participants are using the same computer
- Users issue transactions to call programs on the computer
- Everyone shares the same resources and storage
- The computer has no explicit, single owner
- Using the computer's resources costs money

Election example using a world computer

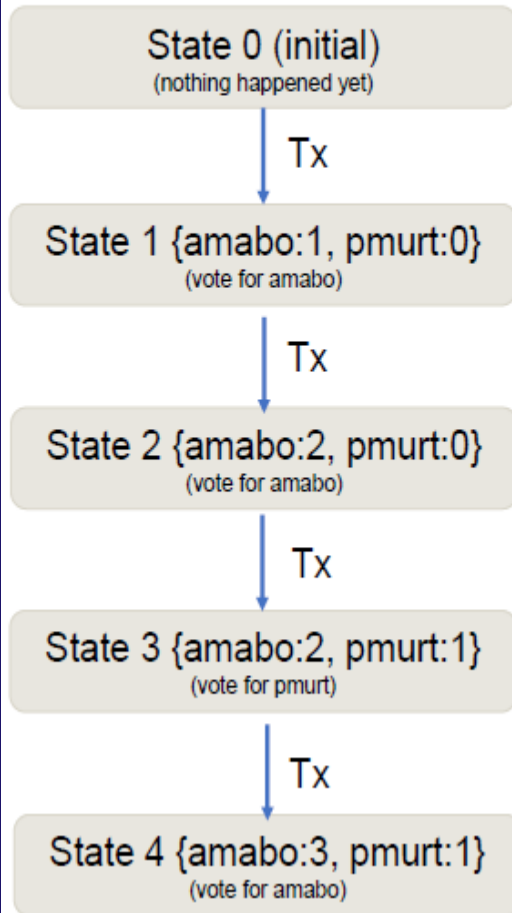


State of the world

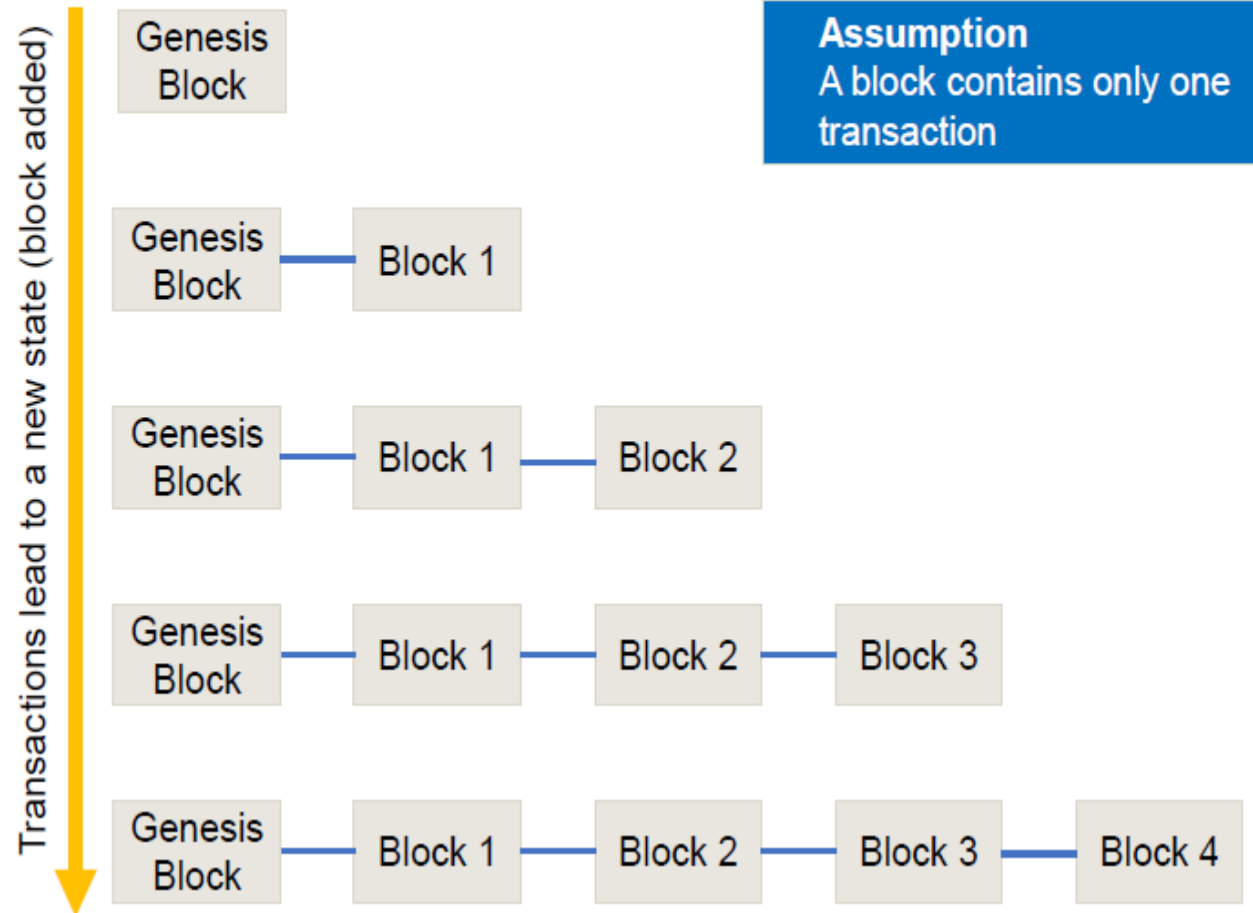


The blockchain as a state machine

State of the world



Blockchain

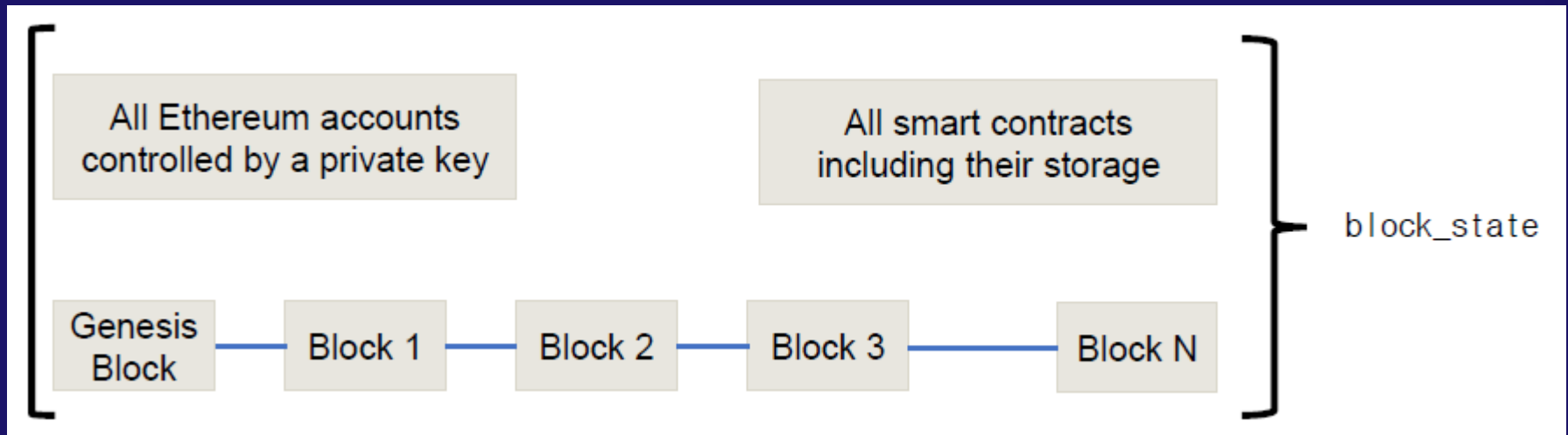


Assumption

A block contains only one transaction

Ethereum

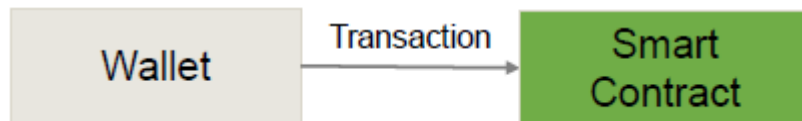
- The EVM specifies an execution model for state changes of the blockchain.
- Formally, the EVM can be specified by the following tuple:
(block_state, transaction, message, code, memory, stack, pc, gas)
- 32 byte instruction word size
- The **block_state** represents the global state of the whole blockchain including all accounts, contracts and storage



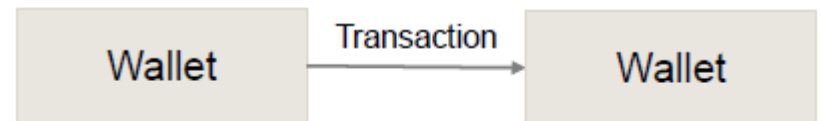
Ethereum

- A **transaction** is a signed data package that is always sent by a wallet and contains the following data field :
 - *Nonce* : The transaction number for this account, starting with 0.
 - *To* : The address of the recipient of the message
 - *Value* : The amount of **wei** (ether) to transfer from the sender to the recipient
 - *Gas Price* : The price in wei that this transaction will pay for Gas
 - *Gas Limit* : The maximum amount of gas that this transaction can consume.
 - *Data* : Data that the code in the EVM processes
 - *Signature* (v, r, s) : A signature identifying the sender
- There are two types of transactions: From wallet to wallet and from wallet to smart contract

Type 1: Wallet to Smart Contract

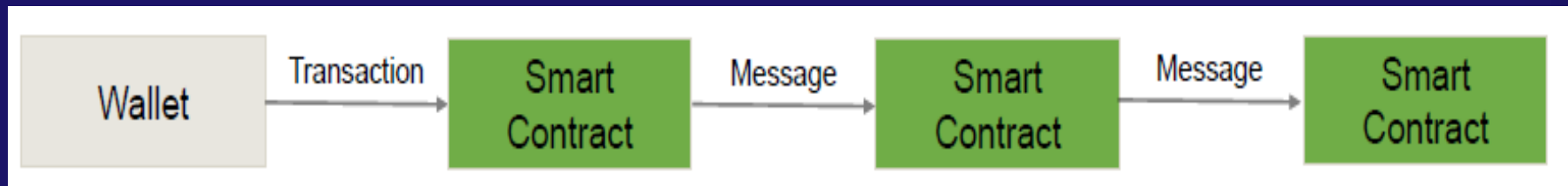


Type 2: Wallet to Wallet

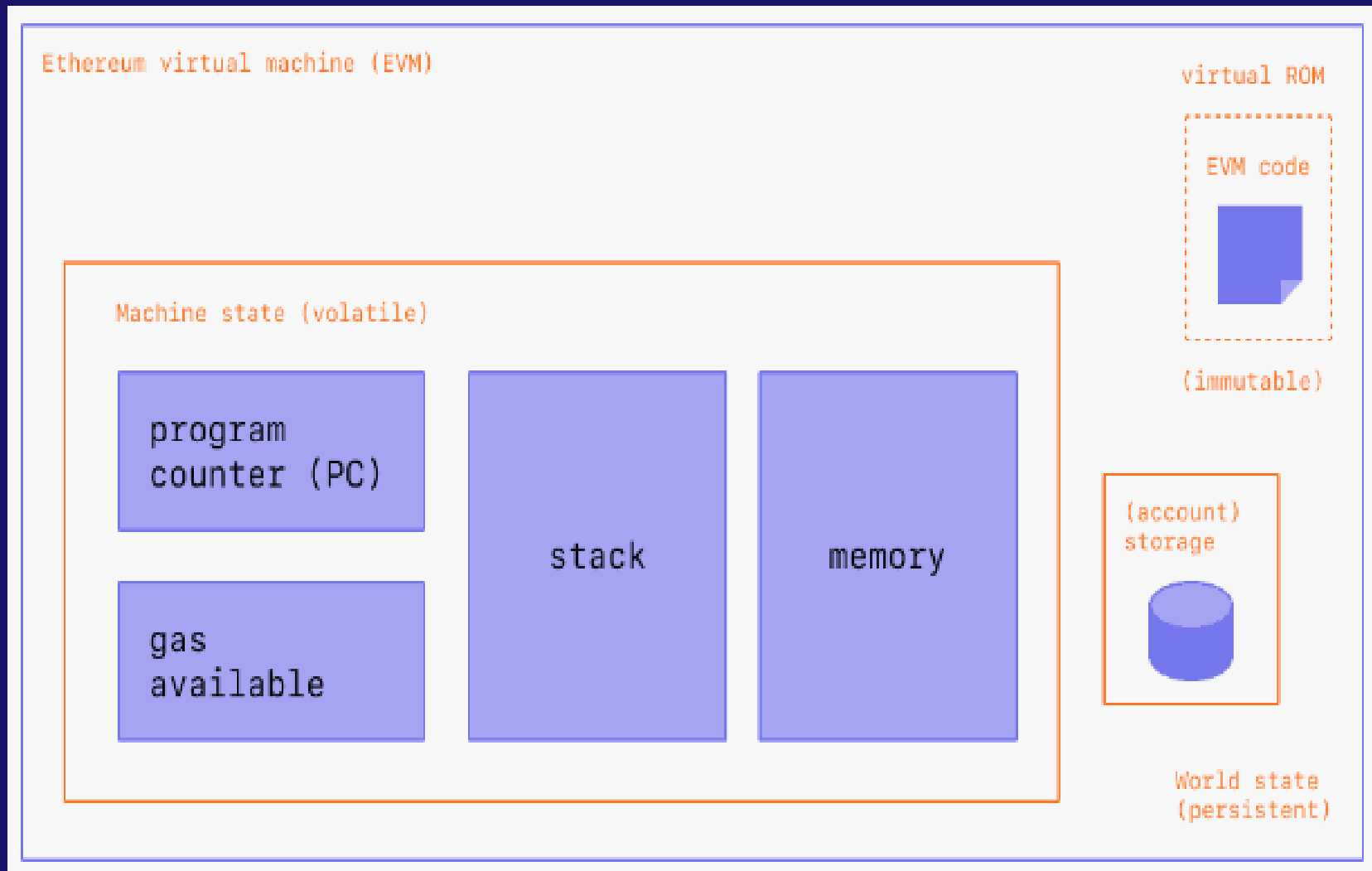


Ethereum

- A **message** is very similar to a transaction. Messages are only sent by contracts and exist only virtually, i.e. they are not mined into a block like transactions. It contains :
 - *The sender of the message (implicit)*
 - *The recipient of the message*
 - *The amount of ether to transfer alongside the message*
 - *An optional data field*
 - *A Gas limit value*
- Whenever a contract calls a method on another contract, a virtual message is sent. Whenever a wallet calls a method on a contract, a transaction is sent.



Ethereum : EVM



Ethereum : EVM

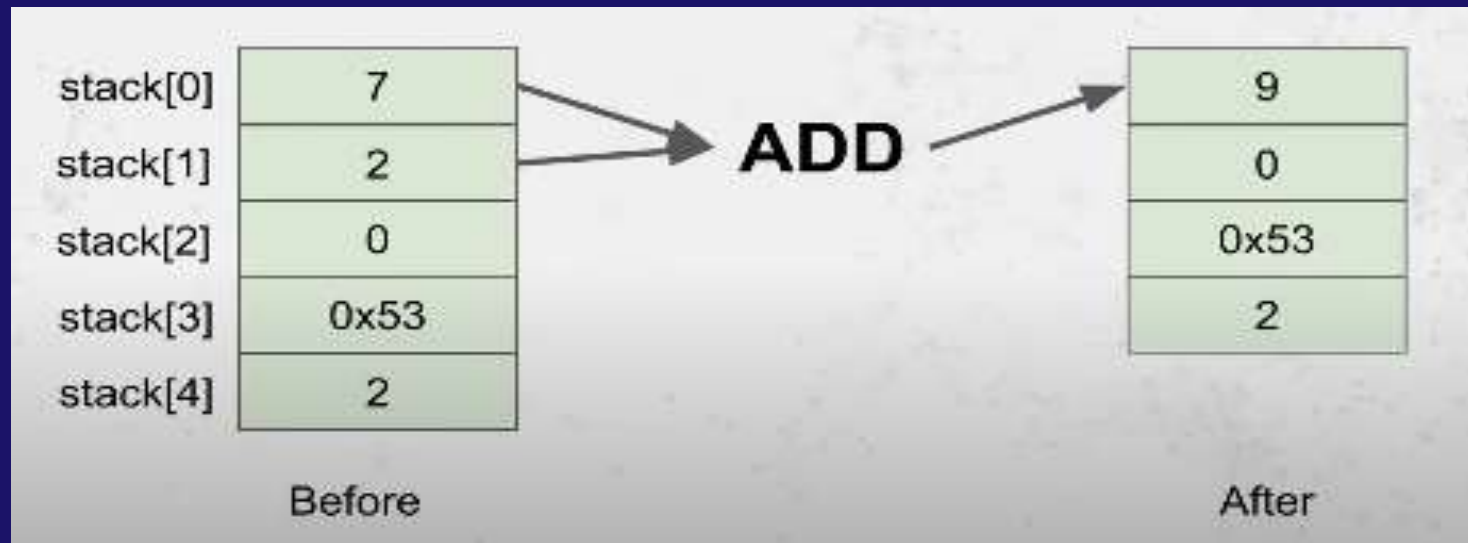
- The EVM makes the code portable across different machines.
- The EVM is a stack-based processor.
- The EVM has 140 opcodes which allow it to be Turing complete.
- Each opcode takes 1 byte of storage space.
- The size of the stack or stack depth is 1024 items, and each item is 256 bit word with a total of 16 slots.
- It also has a contract memory (non-persistent) for complicated operations.
- Reading from storage is free, but writing to storage is extremely expensive.

Ethereum : EVM

- **Code** : The code basically represents a smart contract as bytecode. For the EVM, a smart contract is a sequence of **opcodes** similar to assembly code.
- *Example :*
 - PUSH1 0x60*
 - PUSH1 0x40*
 - MSTORE*
 - PUSH1 0x04*
 - CALLDATASIZE*
 - LT*
 - PUSH2 0x00b6*
 - JUMPI*
 - PUSH4 0xffffffff*
- **Memory** : An infinitely expandable byte array that is non-persistent and used as temporal storage during execution.

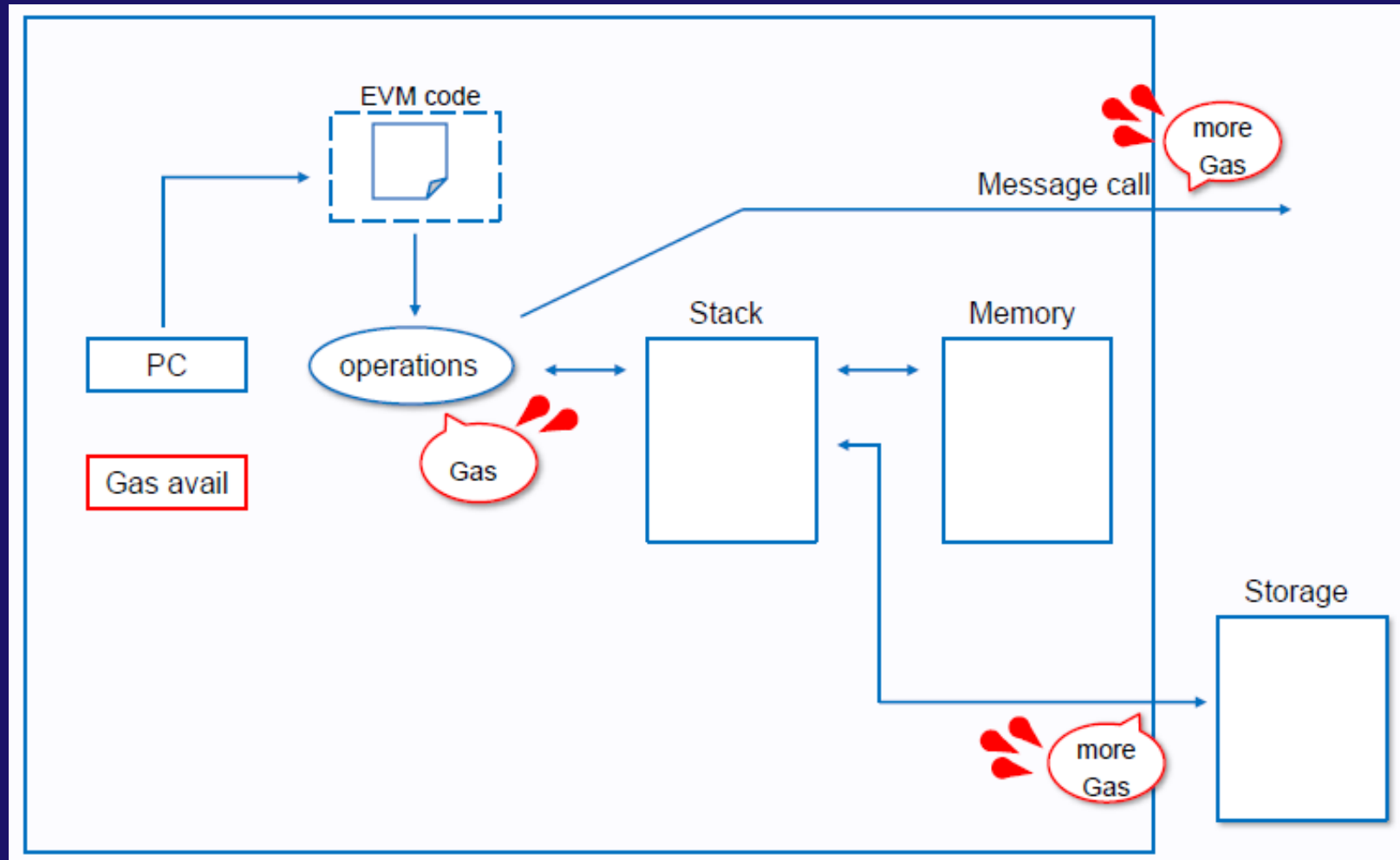
Ethereum : EVM

- The **stack** is also used as a fast, non-persistent buffer to which 32 byte (256 bit) values can be pushed and popped during execution.



- The **program counter PC** is always initialized with 0 and points to the position of the current opcode instruction.

Ethereum : Gas



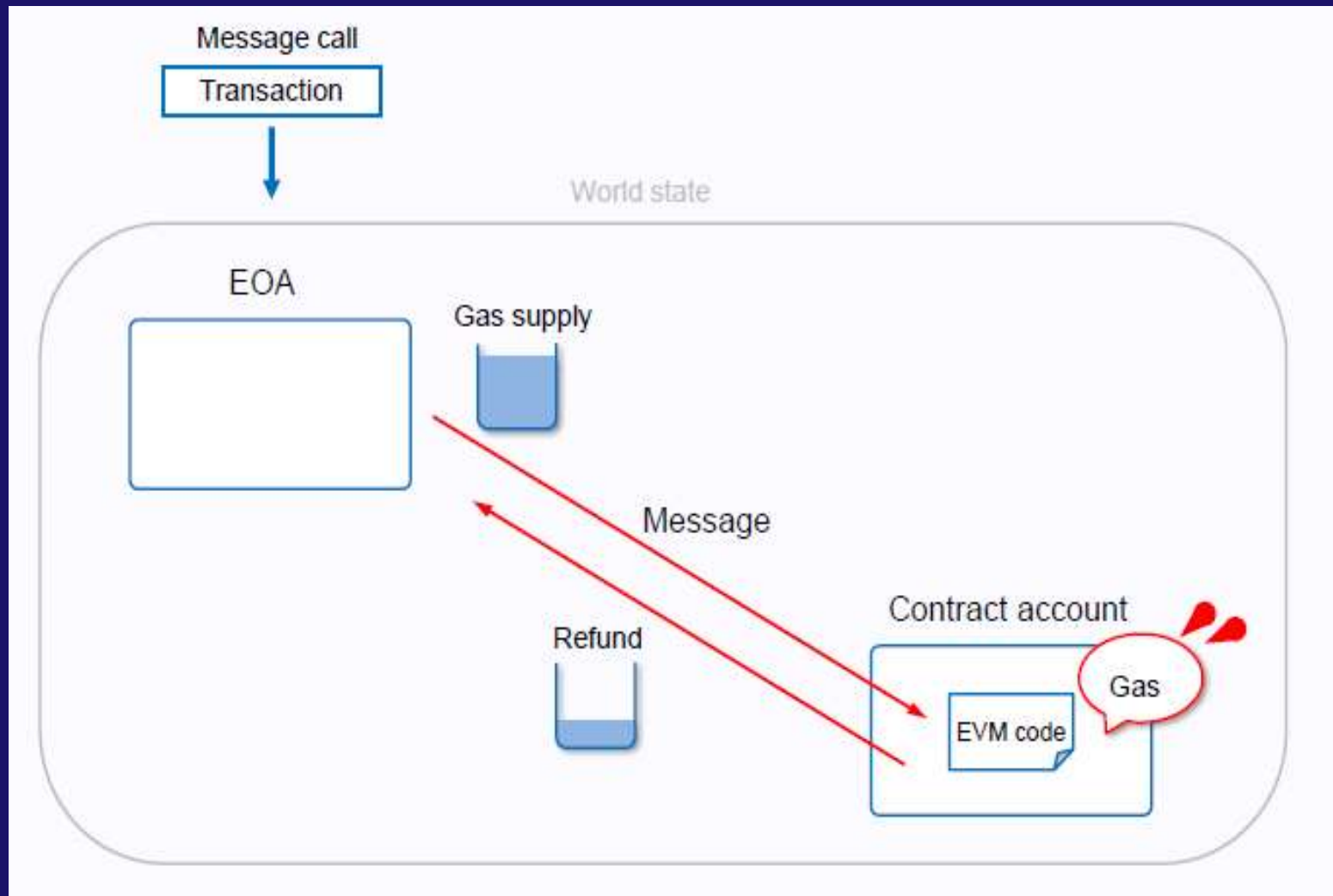
Ethereum : Gas

- Every executed opcode instruction uses a miner's computational resources and therefore costs a certain fee (called gas).
- Each opcode uses a certain amount of gas which may depend on the arguments of the operation, e.g., number of bytes to be allocated.
- At the beginning of a transaction the sender must specify a maximum amount of gas that he is willing to pay for the transaction to be executed.
- The sender can set an arbitrary amount of Ether he is willing to pay for each instruction called gas price.
- The final costs for each transaction are used $\text{gas} * \text{gas price}$.
- If a transaction requires more gas as the maximum specified gas, the transaction will fail. On the other hand, if it takes less, the sender only pays the gas that was used.

Ethereum : Gas

Opcode	Name	Description	Gas
`0x00`	STOP	Halts execution	0
`0x01`	ADD	Addition operation	3
`0x02`	MUL	Multiplication operation	5
`0x03`	SUB	Subtraction operation	3
`0x04`	DIV	Integer division operation	5
`0x05`	SDIV	Signed integer division operation (truncated)	5
`0x06`	MOD	Modulo remainder operation	5
`0x07`	SMOD	Signed modulo remainder operation	5
`0x08`	ADDMOD	Modulo addition operation	8
`0x09`	MULMOD	Modulo multiplication operation	8
`0x0a`	EXP	Exponential operation	10*
`0x0b`	SIGNEXTEND	Extend length of two's complement signed integer	5
`0x0c` - `0x0f`	Unused	Unused	
`0x10`	LT	Less-than comparison	3
`0x11`	GT	Greater-than comparison	3
`0x12`	SLT	Signed less-than comparison	3
`0x13`	SGT	Signed greater-than comparison	3
`0x14`	EQ	Equality comparison	3
`0x15`	ISZERO	Simple not operator	3

Ethereum : Gas and Fee



Ethereum : Account types

- Ethereum supports two types of accounts identified by 32-byte address : *Externally Owned Account (EOA)* and *Smart Contract Account*

Private key

64 hexadecimal character/32 bytes/256 bits



ECDSA

ECDSA Public key

128 hexadecimal character/64 bytes/512 bits



Keccak-256

Hashed public key

64 hexadecimal character/32 bytes/256 bits



Pruning

Account address

40 hexadecimal character/20 bytes/160 bits

699EE77F6467211CDD03F4012B6FEA9377EBD34F107D18C5509CE6AD85113C00

Private Key

EC Derivation SECP256K1

B38317AC43DE95201EA98A0A07990770B1327A7C14E77D9CBA3C922A83758BA5
5CCCE77DC93F7EC936E13BCD4E78552B17C08E83030624A258ADC29759BC19F5

Public Key

Keccak-256

8BC5CBDEB3E3A4D8E4C8AC5A99FDD90FF61DD08CF049155D18E086F7806641B

Ethereum Address

Ethereum : Account types

■ Externally Owned Account

- *Owned by some external entity (person, corporation, etc.)*
- *Accounts that are controlled by a ECDSA private key do not have any code stored on the blockchain.*
- *Can be seen as the default wallet of a user.*
- *It can sign transactions, issue smart contract functions calls and send Ether from one account to another.*
- *The origin of any transaction is always an account controlled and signed by the private key.*

■ Smart Contract Account

- *Account controlled by the contract code.*
- *Contracts can send messages to other accounts, both externally and smart contracts.*
- *Code execution triggered by transactions or function calls (msg)*

Ethereum : Account types

- On an abstract level, an Ethereum account is a 4-tuple containing the following data : (nonce, balance, contract_code, storage)

nonce

An increasing number that is attached to any transaction to prevent replay attacks and double spending.

balance

The current account balance of the account in Ether.

contract_code

The bytecode representation of the account. If no contract code is present, then the account is externally controlled.

storage

The data storage used by the account and empty by default. Only contract accounts can have their own storage.

Ethereum : Account types

